CUSTOMER SHOPPING BEHAVIOR ANALYSIS

A Data Analytics Project

An end-to-end analysis using

Python | PostgreSQL | Power BI

Project Overview

The project examines customer shopping behaviour using a dataset containing 3,900 purchase records and 18 attributes, including customer demographics, purchase items, product categories, purchase amounts, review ratings, discounts, subscription status, shipping methods and previous purchases. The objective is to uncover insights into spending patterns, customer segments, product popularity, customer demographics, subscription behavior, impact of discounts or promotions, and seasonal influences to help the company make informed decisions, optimize product offerings and enhance customer satisfaction.

Dataset Summary

- Rows: 3,900
- Columns: 18
- Dataset Glossary:

Key Features	Column Name	Description
Customer Demographics	Customer ID	Unique identifier for each customer
	Age	Age of the customer
	Gender	Gender of the customer (Male/Female)
	Location	Region or store where the item was purchased
	Subscription Status	Whether the customer has a subscription (Yes/No)
Purchase Details	Item purchased	Name of the purchased item
	Category	Type of the purchased item
	Purchase Amount (USD)	Amount spend to purchase the item, in USD
	Size	Size of the purchased item
	Color	Color of the purchased item
	Season	Season in which the item was purchased
Shopping Behavior	Review Rating	Customer's rating of the product
	Shipping Type	Method of shipping selected
	Discount Applied	Whether the customer has applied a discount (Yes/No)
	Promo Code Used	Whether the customer has used a promo code (Yes/No)
	Previous Purchases	Number of previous purchases made by the customer
	Payment Method	Method of payment used by the customer
	Frequency of purchases	How frequently the purchases were made by the customer (Weekly, Fortnightly, Monthly)

Exploratory Data Analysis with Python

The project begins with data preparation and cleaning using python. The following steps were performed:

- **Data Loading:** Imported dataset using Pandas.
- **Initial exploration:**
  - Used df.head() to preview the dataset
  - Used df.info() to understand the structure of the dataset

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Customer ID             3900 non-null   int64  
 1   Age                     3900 non-null   int64  
 2   Gender                  3900 non-null   str    
 3   Item Purchased          3900 non-null   str    
 4   Category                3900 non-null   str    
 5   Purchase Amount (USD)   3900 non-null   int64  
 6   Location                3900 non-null   str    
 7   Size                    3900 non-null   str    
 8   Color                   3900 non-null   str    
 9   Season                  3900 non-null   str    
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   str    
 12  Shipping Type           3900 non-null   str    
 13  Discount Applied        3900 non-null   str    
 14  Promo Code Used         3900 non-null   str    
 15  Previous Purchases      3900 non-null   int64  
 16  Payment Method          3900 non-null   str    
 17  Frequency of Purchases  3900 non-null   str    
dtypes: float64(1), int64(4), str(13)
memory usage: 835.9 KB
```

  - Used df.describe() to see the statistical overview of the dataset

```
df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Previous Purchases	Payment Method	Frequency of Purchases
3900.000000	3900	3900
NaN	6	7
NaN	PayPal	Every 3 Months
NaN	677	584
25.351538	NaN	NaN
14.447125	NaN	NaN
1.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

- **Handling Missing Data:** Identified 37 missing values in Review Rating column and replaced them with the median review rating for the corresponding product category.

- **Duplicate Checks:** Used duplicate-row check, which returned zero exact duplicate rows.

- **Column Standardization:** Renamed columns to **snake case** to improve readability and consistency

- **Feature Engineering:**
    - Created an age_group column to classify customers into five age groups (18-25, 26-35, 36-45, 46-55, 56+)
    - Created purchase_frequence_days column by mapping purchase-frequency labels to approximate day intervals.

- **Data Consistency Check:** Compared discount_applied and promo_code_used columns and removed the redundant promo_code_used column after confirming that both columns contained identical values

- **Database integration:** Connected Python to PostgreSQL loaded the prepared dataset for SQL analysis.

Data Analysis Using SQL (Business Transaction)

Customer Demographics, Segmentation & Behavior

1. **Purchase Amount by Gender:** The total revenue generated by male and female customers was compared. It was observed that male customers contributed a substantially larger share of the recorded purchase value in this dataset. However, the average purchase amount for female customers was slightly higher ($60.25 versus $59.54).

	gender text	total_purchase_amount numeric	average_purchase_amount numeric
1	Male	157890	59.54
2	Female	75191	60.25

2. **Average Customer Age:** The average age of customers was calculated to provide a general overview of the customer population represented in the data. The average age was approximately 44.07 years

	average_customer_age numeric
1	44.07

3. **Purchase Amount by Age Group:** The 56+ group is the largest age band by record count (1,105) and has the highest total purchase amount ($65,256).

	age_group text	number_of_records bigint	total_purchase_amount numeric	average_purchase_amount numeric
1	18-25	571	34630	60.65
2	26-35	742	44342	59.76
3	36-45	729	43234	59.31
4	46-55	753	45619	60.58
5	56+	1105	65256	59.06

The 56+ age group contributes the most total amount because it contains most records, but its average purchase amount is not highest. The 18-25 and 46-55 group shows slightly higher average amounts in this dataset.

4. **Purchase Amount by Subscription Status:** There are 1053 records marked as subscribed and 2847 marked not subscribed. Subscribers account for $62645 in total purchase amount, compared with $170436 for non-subscribers. However, average amount is $59.49 for subscribers and $59.87 for non-subscribers.

	subscription_status text	total_customers bigint	avg_purchase_amount numeric	total_purchase_amount numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

The subscriber group does not have a significantly higher average purchase amount in this dataset. Non-subscribers generate more total amount partly because there are substantially more non-subscriber records. The data has no subscription fee, lifetime value, churn, etc. measures, so it cannot determine whether subscriptions are profitable or increase spending.

5. Customer Segmentation by Purchase History

	customer_segment text	previous_purchase_range text	Number of Customers bigint	percentage_of_customers numeric
1	New	1-5 purchases	424	10.87
2	Returning	6-10 purchases	360	9.23
3	Loyal	11+ purchases	3116	79.90

Loyal segment accounted for approximately 79.90% of the dataset, while New customers accounted for approximately 10.87%.
Note: Most records belonged to customers with 11 or more previous purchases. The dataset is therefore heavily weighted toward customers with a substantial recorded purchasing history.

6. Purchase amount by customer segment

	customer_segment text	number_of_customers bigint	total_purchase_amount numeric	average_purchase_amount numeric
1	New	424	25692	60.59
2	Returning	360	21872	60.76
3	Loyal	3116	185517	59.54

The average purchase amount across all three segments is relatively similar, ranging from $59.54 to $60.76. Loyal customers generate the highest total purchase amount mainly because this segment contains significantly more customers, rather than because its average purchase amount is higher.

7. Relationship Between Repeated Purchasing and Subscription Adoption

	customer_group text	customer_count bigint	subscribers bigint	subscription_rate_percentage numeric
1	Repeat Buyers: More than 5 Previous Purchases	3476	958	27.56
2	Other Customers: 5 or Fewer Previous Purchases	424	95	22.41

Customers with more than five previous purchases had a higher subscription rate in the dataset. This indicates an association between recorded purchase history and subscription status, but it does not prove that repeat purchasing causes subscription adoption.

Product and Category Performance

8. **Most commonly purchased item:** The most commonly purchased items are Blouse, Pants, Jewelry, with 171 records each.

	item_purchased text	purchase_count bigint
1	Blouse	171
2	Jewelry	171
3	Pants	171

9. **Top 5 products by Average Review Rating:**

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

These products had relatively higher review ratings than other products in the dataset. However, the ratings were not close to the maximum score of five, indicating that there may still be opportunities to improve customer satisfaction.
Note: These averages are based on available ratings after category-median imputation. The small differences should not be treated as statistically significant without further testing.

10. **Total Purchase Amount by Product Category:**

	category text	number_of_records bigint	total_purchase_amount numeric	average_purchase_amount numeric
1	Clothing	1737	104264	60.03
2	Accessori...	1240	74200	59.84
3	Footwear	599	36093	60.26
4	Outerwear	324	18524	57.17

Clothing generated the highest total purchase amount, followed by Accessories, Footwear and Outerwear. Clothing also has the largest record count. Average purchase amounts across categories are relatively close, so the total-revenue ranking appears to be driven substantially by the number of records rather than a large difference in average spend.

11. **Top 3 Most Frequently Purchased Products within Each Category**
Blouse and pants tied for the highest count in clothing. Jewelry was the most frequently recorded product in accessories, while sandals led footwear. Outerwear contained only two distinct products in the dataset i.e., jacket and coat.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

12. Discount usage and spending comparison

	discount_applied text	number_of_customers bigint	percentage_of_customers numeric	total_purchase_amount numeric	average_purchase_amount numeric
1	Yes	1677	43.00	99411	59.28
2	No	2223	57.00	133670	60.13

The analysis shows that 43% of customers used discounts, while 57% made purchases without them. Customers who used discount contributed a lower total purchase amount than customers who did not, although the latter group also contained more customers. The average purchase amount was slightly higher among customers who did not use discounts. These findings describe an association between discount usage and purchase amounts but do not establish a causal relationship.

13. Top 5 Products by Percentage of Discounted Purchases

	item_purchased text	total_purchases bigint	discounted_purchases bigint	discount_usage_percentage numeric
1	Hat	154	77	50.00
2	Sneakers	145	72	49.66
3	Coat	161	79	49.07
4	Sweater	164	79	48.17
5	Pants	171	81	47.37

Hat had the highest discount usage percentage, with discounts appearing in 50.00% of its records, followed by Sneakers and Coat.
These products had relatively high proportions of discounted records. This indicates where discounts were most frequently recorded, not necessarily where discounts generated the highest sales or profit.

14. **Discounted Purchases Above the Overall Average**

A total of 839 customer records met both conditions: a discount was applied, and the purchase amount was greater than the overall average purchase amount. They account for approximately 21.51% of all the customers.

The business could investigate the products and customer segments represented in these records to understand where discounts coincide with higher purchase values. However, these records alone do not prove that discounts caused customer to spend more.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91

Total rows: 839 Query complete 00:0(

15. **Customers with subscription using discount**

	subscribed_customers_using_discount bigint
1	1053

All 1053 customers with a subscription status of “Yes” also had a discount applied in the dataset. This pattern suggests a possibility that discounts could be a subscription benefit, for which further investigation is needed.

Sales Operations & Trends

16. Most Commonly Used Payment Methods

	payment_method text	payment_count bigint
1	PayPal	677
2	Credit Card	671
3	Cash	670
4	Debit Card	636
5	Venmo	634
6	Bank Transfer	612

PayPal is the most common payment method with 677 records, narrowly ahead of Credit Card and Cash. The top methods are close in count, so the data does not suggest that one method overwhelmingly dominates.
Note: The business could use this result to ensure that the leading methods remain available, while considering transaction fees and customer experience before making payment-policy changes.

17. Most Common Shipping Type Among Customers with Review Ratings Above 4

	shipping_type text	customer_count bigint
1	Standard	271

Among customer records with a review rating above 4, Standard Shipping was the most common shipping type, appearing in 271 records.

18. Purchase Amount by Season

	season text	number_of_records bigint	total_purchase_amount numeric	average_purchase_amount numeric
1	Fall	975	60018	61.56
2	Spring	999	58679	58.74
3	Winter	971	58607	60.36
4	Summer	955	55777	58.41

Fall generated the highest total purchase amount, at $60,018, and also had the highest average purchase amount, at approximately $61.56. Summer recorded lowest total and average purchase amounts.
Note: Since the data does not include the transactional dates or multiple years of observations, so it cannot establish a recurring seasonal trend.

19. Clothing Purchase Amount by Size

	size text	number_of_records bigint	total_purchase_amount numeric	contribution_percentage numeric
1	M	778	47041	45.12
2	L	481	27864	26.72
3	S	284	17416	16.70
4	XL	194	11943	11.45

Size M has the greatest number of records and the largest purchase amount, contributing the most to the total clothing purchase amount (45.12%), followed by L (26.72%), S (16.70%) and XL (11.46%).

20. Purchase Amount by Location

The customer data includes 50 distinct locations. The results show that Montana leads in total purchase amount, followed by Illinois and California. However, locations with the highest total purchase amounts did not necessarily have the highest average purchase amounts. For example, Alaska and Pennsylvania had higher purchase amounts than Montana, but their total purchase amounts are lower because they have fewer customers.

	location text	customer_count bigint	total_purchase_amount numeric	average_purchase_amount numeric
1	Montana	96	5784	60.25
2	Illinois	92	5617	61.05
3	California	95	5605	59.00
4	Idaho	93	5587	60.08
5	Nevada	87	5514	63.38
6	Alabama	89	5261	59.11
7	New York	87	5257	60.43
8	North Dakota	83	5220	62.89
9	West Virginia	81	5174	63.88
10	Nebraska	87	5172	59.45
11	New Mexico	81	5014	61.90
12	Minnesota	88	4977	56.56
13	Pennsylvania	74	4926	66.57
14	Mississippi	80	4883	61.04
15	Alaska	72	4867	67.60
16	Vermont	85	4860	57.18

Total rows: 50 Query complete 00:00:00.130

Dashboard in Power BI







Business Recommendations

- **Prioritize the leading categories and popular products:** Clothing and Accessories together contribute about 76.6% of record purchase value. The business could investigate demand for products under these categories when planning inventory and promotional campaigns. Product profitability, stock availability, and return rates should also be considered before prioritizing inventory.

- **Age profile:** The 56+ band is the largest age group and has the highest total amount, but not the highest average amount. The business may wish to investigate whether this reflects a larger sample, customer mix, or different product preferences.

- **Payment options:** PayPal is the most common method, but several payment methods have similar counts, suggesting that supporting multiple payment methods may be important. The business should ensure that commonly used options remain convenient and reliable, and also review costs and failed-payment rates.

- **Improving subscription adoption:** The business could investigate why customers choose not to subscribe and evaluate whether clearly communicated subscription benefits or targeted offers could improve subscription adoption.

- **Evaluate Subscription performance:** Subscribers do not have higher average purchase amounts than non-subscriber. The business should evaluate subscription performance using additional measures, such as retention, purchase frequency, lifetime value, and profit margin. These measures would help determine whether subscriptions provide meaningful business benefits beyond the average purchase amount.

- **Evaluate Discount Effectiveness:** The business could investigate whether discounts are associated with higher purchase amounts or increased customer engagement. Discount

effectiveness should ultimately be evaluated using incremental sales and profit margins rather than discount usage alone.

- **Seasonality:** Fall leads recorded seasonal amount, though differences are not large. The business could investigate which customer groups and products contribute to Fall's higher purchase amount. Using historical data would help in validating whether seasonal patterns are consistent enough to guide inventory and marketing plans.

- **Ratings and shipping:** Standard shipping is most common among high-rated records. However, the business needs to compare the proportion of high ratings across shipping types to determine if shipping method influences customer satisfaction.

- **Use customer segments carefully:** The business could investigate ways to retain customers in the Loyal segment while developing onboarding and engagement strategies for customers with fewer previous purchases.

Limitations

- The analysis is descriptive and does not establish causation.
- The dataset does not establish whether each row represents a unique customer, an individual transaction, or a customer-level purchase record.
- The 37 missing customer review ratings were imputed using category-specific medians, which affect average review rating calculations.
- The data does not include transaction dates, product costs or profit margins. Therefore, the analysis describes recorded purchase amounts rather than verified revenue across multiple transactions.

Conclusion

The project demonstrates an end-to-end data analytics workflow, from data inspection and cleaning in Python to business-focused analysis using SQL and visualization through a Power BI dashboard.

The analysis of 3,900 customer records identified Clothing as the with highest purchase amount. A tie was seen among Blouse, Pants, and Jewelry as the most commonly purchased item, while PayPal was the most commonly used payment method. Customers aged 56 and above contributed the largest share in total purchase amount, among the selected age groups. It also highlighted opportunities to examine discount effectiveness, subscriber behavior, seasonal variation, and customer retention in more depth. Overall, this project showcases how data analytics help business to analyze customer shopping behavior and generate insights that support data informed decision-making.